

SETS

- $\in$ element of
- $\notin$ not element of
- $\emptyset$ empty set
- $\mathbb{U}$ universal set
- $\subseteq$ Subset $\subset$ Proper subset
- $\nsubseteq$ not
- $\cup$ - union
- $\cap$ intersect

INDICES RULES

- $a^m \times a^n = a^{m+n}$
- $a^m \div a^n = a^{m-n}$
- $(a^m)^n = a^{m \times n}$
- $a^{\frac{1}{m}} = \sqrt[m]{a}$
- $a^{\frac{n}{m}} = \sqrt[m]{a^n}$
- $a^{-m} = \frac{1}{a^m}$
- $a^0 = 1$
- $a^1 = a$

SIMPLE & COMPOUND INTEREST

- $\frac{P \times r}{100} = \text{Simple interest}$
- $P(1 + \frac{r}{100})^n = \text{Compound increase}$
- $P(1 - \frac{r}{100})^n = \text{Compound decrease}$

CONVERSIONS

- $m^3 \xrightarrow{\times 1000} l \xrightarrow{\times 1000} cm^3$
 $\div 1000 \quad \div 1000$
- $\text{mile} \xrightarrow{\times \frac{8}{5}} km$
 $\times \frac{5}{8}$
- $km/h \xrightarrow{\times \frac{5}{18}} m/sec$
 $\times \frac{18}{5}$

PERCENTAGES

- Profit percentage = $\frac{SP - CP}{CP} \times 100$
 - Loss Percentage = $\frac{CP - SP}{CP} \times 100$
- SP = Selling Price
CP = Cost price

maths

ALGEBRA

- $(a+b)^2 = a^2 + 2ab + b^2$
- $(a-b)^2 = a^2 - 2ab + b^2$
- $(a+b)(a-b) = a^2 - b^2$
- $(a+b)^2 - 2ab = a^2 + b^2$

MENSURATION

- $C^2 = a^2 + b^2$ — Pythagoras theorem
- Area of Parallelogram = $b \times h$
- Area of trapezium = $\frac{1}{2}(a+b)h$
- Area of Sector = $\pi r^2 \times \frac{\theta}{360}$
- Arc length = $2\pi r \times \frac{\theta}{360}$
- Volume of cylinder = $\pi r^2 h$
- Volume of cone = $\frac{1}{3}\pi r^2 h$
- Volume of sphere = $\frac{4}{3}\pi r^2$
- Volume of pyramid = $\frac{1}{3} \text{base area} \times \text{height}$

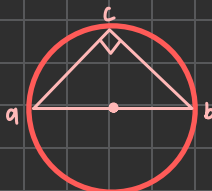
Circle theorems



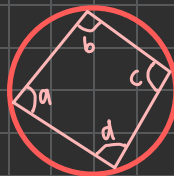
angle at centre from same arc length.



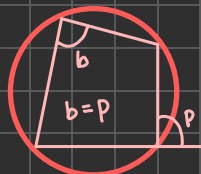
angles in same segment / from same arc length



Angles in Semicircle
 $\angle acb = 90^\circ$



$a + c = 180$
 $b + d = 180$
opposite angles supplementary in cyclic quadrilateral

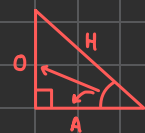


Alternate Segment theorem



Tangents to circle theorem

TRIGONOMETRY



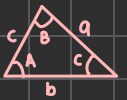
SOH $\sin \theta = \frac{O}{H}$
CAH $\cos \theta = \frac{A}{H}$
TOA $\tan \theta = \frac{O}{A}$

Area of triangle:

- $\frac{1}{2} b \times h$
- $\frac{1}{2} ab \sin c$

SINE RULE

$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$



COSINE RULE

- $a^2 + b^2 - 2ab \cos C$
- to find length of side c^2
- $\cos^{-1} \left(\frac{a^2 + b^2 - c^2}{2ab} \right)$
- to find angle C

STRAIGHT LINE GRAPHS

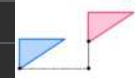
Gradient: $\frac{y_2 - y_1}{x_2 - x_1}$
midpoint: $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
length: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

TRANSFORMATIONS

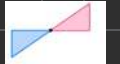
Reflection



3. Translation e.g.

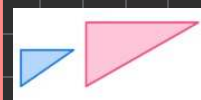


Rotation



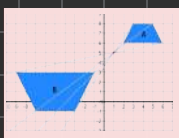
- Centre of rotation
- draw lines & see where perpendicular bisectors connect
- angle & direction
- anti clockwise (↺) clockwise (↻)

Enlargement



- Scale factor
- centre of rotation

Negative enlargement



CUMULATIVE FREQUENCY

Upper quartile = $\frac{3}{4} (\text{total frequency})$
lower quartile = $\frac{1}{4} (\text{total frequency})$

upper bound

lower bound

- + UB + UB
- UB - LB
- × UB × UB
- ÷ UB ÷ LB
- + LB + LB
- LB - UB
- × LB × LB
- ÷ LB ÷ UB